Eunseong Choi, Ph.D.

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Summary

I'm a Research Scientist at Liner. My research lies in **Information Retrieval** and **Natural Language Processing**, with an emphasis on building **robust and efficient Retrieval-Augmented Generation** frameworks. My work has received 300+ citations. Representative research includes:

- **Retrieval-Augmented Generation:**
  – Context–memory conflicts [C1]
  – Prompt compression for efficient LLMs [C4]
- **Information Retrieval:**
  – Passage re-ranking [C2]
  – Learned sparse representation [C8]
  – Generative recommendation [C3]
- **Natural Language Processing:**
  – Metaphor detection [C10]
  – Query reduction [C7]
- **Machine Learning:**
  – Knowledge tracing [C6]
  – Extreme multi-label classification [C9]

Experience

2025.12 – Current
ML Research Scientist, Core Tech Division, Liner
• Trained and deployed a neural re-ranker, improving retrieval performance by 12.5% over the previous production model
• Built a daily incremental pipeline to improve search index freshness
• Building an LLM routing system using real-world user queries

2025.3 – 2025.4
Research Intern, Search & Ranking Modeling, NAVER
• Worked on continual fine-tuning for recency-aware dense retrieval

2021.7 – 2021.8
Research Intern, Search CIC, NAVER
• Developed a sparse retrieval method for efficient first-stage retrieval

Education

2020.3 – 2026.2
M.S./Ph.D., *Artificial Intelligence*, Sungkyunkwan University
Advisor: Prof. Jongwuk Lee
Thesis: *Improving Evidentiality and Compression for Retrieval-Augmented Generation*

2012.3 – 2020.2
B.S., *Architecture*, Sungkyunkwan University
B.S., *Samsung Convergence Software Course*, Sungkyunkwan University

Publications

[C1] Conflict-Aware Soft Prompting for Retrieval-Augmented Generation *EMNLP 2025*

Eunseong Choi, June Park, Hyeri Lee, Jongwuk Lee
• Mitigated context–memory conflicts in retrieval-augmented generation using adversarial soft prompting

[C2] Multi-view-guided Passage Reranking with Large Language Models *EMNLP 2025*
Jeongwoo Na, Jun Kwon, **Eunseong Choi**, Jongwuk Lee
• Proposed efficient and robust LLM-based passage reranker using multi-view embeddings

[C3] GRAM: Generative Recommendation via Semantic-aware Multi-granular Late Fusion *ACL 2025*
Sunkyung Lee, Minjin Choi, **Eunseong Choi**, Hye-young Kim, Jongwuk Lee
• Efficiently leveraging multi-granular semantic information for sequential recommendation

[C4] From Reading to Compressing: Exploring the Multi-document Reader for Prompt Compression *EMNLP 2024 Findings*
Eunseong Choi, Sunkyung Lee, Minjin Choi, June Park, Jongwuk Lee
• Prompt compression using cross-attention, reducing prompt length by 80% while preserving global context

[C5] Multi-granularity Guided Fusion-in-Decoder *NAACL 2024 Findings*
Eunseong Choi, Hyeri Lee, Jongwuk Lee
• Improved open-domain QA by leveraging evidence at multiple levels of granularity

[C6] Forgetting-aware Linear Bias for Attentive Knowledge Tracing *CIKM 2023 (short paper)*
Yoonjin Im*, **Eunseong Choi***, Heejin Kook, Jongwuk Lee
• Modeled forgetting behavior with a linear bias in attention-based models

[C7] ConQueR: Contextualized Query Reduction using Search Logs *SIGIR 2023 (short paper)*
Hye-young Kim*, Minjin Choi*, Sunkyung Lee, **Eunseong Choi**, Young-In Song, Jongwuk Lee
• Query reduction by combining core term extraction and sub-query selection from real-world search logs

[C8] SpaDE: Improving Sparse Representations using a Dual Document Encoder for First-stage Retrieval *CIKM 2022*
Eunseong Choi*, Sunkyung Lee*, Minjin Choi, Hyeseon Ko, Young-In Song, Jongwuk Lee
• Improved sparse retrieval by jointly learning term weighting and semantic expansion with a dual encoder

[C9] Long-tail Mixup for Extreme Multi-label Classification *CIKM 2022 (short paper)*
Sangwoo Han, **Eunseong Choi**, Chan Lim, Hyunjung Shim, Jongwuk Lee
• Addressed label sparsity in extreme classification with mixup-based augmentation

[C10] MelBERT: Metaphor Detection via Contextualized Late Interaction using Metaphorical Identification Theories *NAACL 2021*
Minjin Choi, Sunkyung Lee, **Eunseong Choi**, Heesoo Park, Junhyuk Lee, Dongwon Lee, Jongwuk Lee
• Metaphor detection using contextualized word representations and linguistic theories

Academic Service

Reviewer	ARR 2025 May (EMNLP 2025)
External Reviewer	EMNLP 2022; ACL 2023; SIGIR 2023–2024; KDD 2022, 2026; WSDM 2022; ARR 2025 Feb (ACL 2025)

Honors & Awards

2024 **1st Prize, Best Graduate Research Paper Award**, Sungkyunkwan University.
Presented research on prompt compression for efficient LLM

2023 **2nd Place (Minister's award), AI Grand Challenge for Policy Support AI**, IITP.
Developed a multi-hop retriever and re-ranker for generating policy-supportive documents

Honors & Awards (continued)

2022 **1st Place (Minister's award), AI Grand Challenge for Math Word Problem Solving**, IITP.
Developed a framework for math word problem solving using domain-specific operations

2021 **2nd Prize, Best Graduate Research Paper Award**, Sungkyunkwan University.
Presented research on efficient sparse retrieval using term expansion and re-weighting

Scholarships

2020 – 2022 **Graduate School Scholarship**, Department of AI, Sungkyunkwan University

2019 **Academic Excellence Scholarship**, Sungkyunkwan University

2017 – 2019 **National Scholarship for Science and Engineering Students**, Korea Student Aid Foundation

2015 – 2016 **Academic Excellence Scholarship**, Sungkyunkwan University.

Teaching Assistant

Regular Courses

Spring 2023 **Introduction to Recommender Systems**

Fall 2022 **Fundamentals of Machine Learning**

Fall 2021 **Deep Neural Networks**

Spring 2020 – 2022 **Introduction to Database**

Extracurricular Courses

Machine Learning
LG Electronics May 2020, Oct 2020, May 2021, Oct 2021, Oct 2022, Oct 2024
Samsung SDS Jun 2020
SK Innovation Sep 2020
Machine learning basics, including regression and neural networks

NLP Project
Samsung SDS Sep 2022
Covered topic classification and machine reading comprehension

Industry–Academia Projects

2025.09 – 2026.02 **WeZON**
NL2SQL-based Intelligent Question Answering System Development
Project Manager - Developed an NL2SQL system enabling non-experts to query databases in natural language

2021.07 – 2022.06 **NAVER**
Document Retrieval via Pre-trained Language Models and Inverted Index
Team Leader - Designed a dual-document encoder for effective sparse representations

2020.05 – 2021.04 **NAVER**
Topic-aware Query–Document Matching with Deep Neural Networks
Leveraged VAE-based topic modeling to represent query–document semantics

References

Available upon request